

JIJNASH KUMAR MUKKA

Anakapalli, Andhra Pradesh, India | +91- 86882 11679 | jiijnashkumar@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) |

AI/ML undergraduate engineer with a **9.52/10 CGPA** and hands- on experience across **machine learning, deep learning, Generative AI, AI agents, multimodal AI, computer vision, API development, and cloud- based AI systems**. Research experience at IIIT Nagpur and NIT Calicut, with practical exposure to LLMs and AI agents at IIT Jammu. Experienced in developing ML analysis pipelines, FastAPI- based machine learning services, simulation- based intelligent systems, and healthcare AI applications. Strong foundation in Python, C++, scikit- learn, NumPy, Pandas, FastAPI, REST APIs, Git/GitHub, Google Cloud, and Vertex AI.

EDUCATION

B.Tech, Artificial intelligence & machine (2023 – Present)
SRM Institute of Science and Technology, Tiruchirappalli, CGPA – **9.52/10**
Graduation year (2027)

12th (Senior Secondary Examination) May 2020 – March 2022
SV Defense Academy, Tirupathi, Andhra Pradesh | **76.8%**

10th (Secondary Examination) March 2019 – March 2020
Cambridge Space School, Kasimkota, Anakapalli, Andhra Pradesh | **99%**

Research Experience

Research Intern - Indian Institute of Information Technology Nagpur Dec 2025 – Feb 2026

- Independently worked on multimodal Parkinson's disease detection using voice, facial, and hand/motor biomarkers.
- Developed analysis pipelines for acoustic features including MFCCs, pitch, jitter, shimmer, and HNR using librosa and Parselmouth.
- Applied MediaPipe for facial landmark analysis and explored hand/movement analysis using DeepLabCut.
- Explored Velocity Fatigue Slope (VFS) and confidence- weighted late fusion for combining complementary biomarker information.
- Applied statistical analysis and model evaluation techniques to assess individual modalities and multimodal approaches.

Artificial Intelligence Intern - Indian Institute of Technology Jammu Dec 2025 – Mar 2026

- Gained hands- on experience with Large Language Models, Generative AI, and AI Agents through an intensive AI internship program.
- Worked with concepts including prompt engineering, reasoning, memory, tool usage, and agentic workflows.
- Explored practical approaches for developing AI applications using LLM- based systems and agent workflows.

Research Intern - National Institute of Technology Calicut May 2026 – July 2026

- Contributed to research on an intelligent multi- agent framework for urban traffic signal optimization under changing traffic conditions.
- Conducted literature analysis covering reinforcement learning, multi- agent reinforcement learning, traffic prediction, risk- aware traffic management, and intelligent decision support.
- Contributed to evaluation review, research- gap identification, technical assignments, and presentation of research findings.
- Collaborated with team members to synthesize technical findings and communicate proposed approaches.

Projects

AI- Powered Medical Diagnosis API

- Developed a Python/FastAPI- based machine learning API for disease prediction using structured clinical numerical data.
- Implemented data validation and REST endpoints for serving machine learning predictions.
- Integrated Google Cloud and Vertex AI for cloud- based model deployment and inference.
- Applied machine learning and API development concepts to build a practical healthcare AI service.

Multimodal Parkinson's Disease Detection

- Developed a multimodal AI framework for Parkinson's disease analysis using voice, facial, and hand/motor biomarkers.
- Applied acoustic feature extraction, facial landmark analysis, hand- movement analysis, and multimodal fusion techniques.
- Explored Velocity Fatigue Slope (VFS) and confidence- weighted fusion for integrating complementary biomarker information.

MedDocAssist - Multimodal Clinical NLP Framework

- Developed a clinical NLP framework for named entity recognition, ICD- 10 coding, clinical summarization, and drug- interaction detection.
- Applied NLP techniques to structure clinical information and support automated healthcare decision- support workflows.

EcoSignal- CP - Carbon Pressure Control for Urban CO₂ Emission Waveform Stabilization

- Developed a SUMO- based traffic simulation and predictive control approach for studying urban CO₂ emission behavior.
- Contributed to the technical core, experimentation, analysis, and research paper.
- Conducted simulation- based experimentation and analyzed emission- control behavior across different traffic conditions.
- Presented the research paper at ICRISSET 2026, Chennai, India, August 2026.

Achievements

- **Top 6 National Rank - Smartathon 2025**, Emergency Brain 360
- **5th Place, Gold Medal & Special Prize - MedAIthon 2025**, Centralized Hospital Analytics Platform
- **Research Paper Presentation - ICRISSET 2026**, EcoSignal- CP: Carbon Pressure Control for Urban CO₂ Emission Waveform Stabilization

Technical Skills

Programming: Python, C++

Machine Learning & Data: Machine Learning, Deep Learning, scikit- learn, NumPy, Pandas, Feature Engineering, Model Evaluation, Statistical Analysis

Generative AI: Generative AI, Large Language Models (LLMs), AI Agents, Prompt Engineering

Computer Vision & Multimodal AI: MediaPipe, DeepLabCut, Multimodal AI

Backend & APIs: FastAPI, REST APIs, API Development, Data Validation, Model Deployment

Cloud & Tools: Google Cloud, Vertex AI, Git, GitHub

Research & Applications: Healthcare AI, Reinforcement Learning, Traffic Simulation, Clinical NLP

CERTIFICATIONS

- Red Hat Certified System Administrator (RHCSA)
- IIT Jammu - Artificial Intelligence: LLMs, Generative AI & AI Agents
- Google Cloud - Generative AI
- AWS Academy - Machine Learning
- ServiceNow - AI

LANGUAGES

English, Telugu, Japanese